11)

```
SQL>CREATE TABLE Customer(
 Cust_id VARCHAR(3) PRIMARY KEY,
 Fname VARCHAR(20) NOT NULL,
  Lname VARCHAR(20),
  Area VARCHAR(2) NOT NULL,
   Phone VARCHAR(10));
```

Table created.

```
SQL> CREATE TABLE Movie(
    Mv_no Number(2) PRIMARY KEY,
     Cust_id VARCHAR(3) REFERENCES Customer(Cust_id),
    Title VARCHAR2(30) NOT NULL,
    Star VARCHAR2(2) NOT NULL,
    Price NUMBER(5) CHECK (Price BETWEEN 100 AND
                                    250));
```

Table created.

```
SQL> INSERT INTO Customer VALUES('A01', Ivan, 'Ross', 'SA', 6125467);
```

1 row created.

```
SQL> INSERT INTO Customer VALUES ('A02', 'Vandana', 'Rao',
                'MU', 5560379);
```

1)

```
SQL> CREATE TABLE Customer(
Cust_id VARCHAR(3) PRIMARY KEY,
Fname VARCHAR(20) NOT NULL,
Lname VARCHAR(20),
Area VARCHAR(2) NOT NULL,
Phone VARCHAR(10));
```

Table created.

```
SQL> CREATE TABLE Movie(
Mv_no Number(2) PRIMARY KEY,
Cust_id VARCHAR(3) REFERENCES Customer(Cust_id),
Title VARCHAR2(30) NOT NULL,
Star VARCHAR2(2) NOT NULL,
Price NUMBER(5) CHECK (Price BETWEEN 100 AND
250));
```

Table created.

```
SQL> INSERT INTO Customer VALUES('A014', Ivan, 'Ross', 'SA', 6125467);
```

1 Row created.

```
SQL> INSERT INTO Customer VALUES ('A024', 'Vandana', 'Rao',
'MV', 5560379);
```

1 row created

SQL> INSERT INO Customer VALUES ('A0

('Jyosta', 'PA', 456089);

1 row created.

SQL> INSERT INTO Customer VALUES ('A04', 'Basu', 'Nerambil

'BA', 612501);

1 row created.

SQL> INSERT INTO Customer VALUES ('A05', 'Ravi', 'Shanidha

NA', NULL);

1 row created.

SQL> INSERT INTO Customer VALUES ('A06', 'Rukmini', 'Aiyar',

'AI', 512574);

1 row created.

SQL> INSERT INTO Movie VALUES (1, 'A02', 'Bloody', 'C',

18 1);

1 row created.

SQL> INSERT INTO Movie VALUES (2, 'A09', 'The Film',

'TC', 200);

1 row created.

SQL> INSERT INTO Movie VALUES (3, 'A01', 'Pretty Woman',

'RA', 151);

1 row created.

Student Name: NIHAL

1 row created

SQL> INSERT INO Customer VALUES ('A03', 'Pramada', 'Jgusta', 'PA', 456038);

1 row created.

SQL> INSERT INTO Customer VALUES ('A04', 'Basu', 'Navindil', 'BA', 612501);

1 row created.

SQL> INSERT INTO Customer VALUES ('A05', 'Ravi', 'Shridha', 'NA', NULL);

1 row created.

SQL> INSERT INTO Customer VALUES ('A06', 'Rukmini', 'Ahen', 'Af', 512574);

1 row created.

SQL> INSERT INTO Movie VALUES (1, 'A02', 'Bloody', 'Jc', 18);

1 row created.

SQL> INSERT INTO Movie VALUES (2, 'A04', 'The Film', 'TC', 200);

1 row created.

SQL> INSERT INTO Movie VALUES (3, 'A01', 'Pretty Woman', 'RA', 151);

1 row created.

SQL > INSERT INTO Movies VALUES (4, 'A06', 'Home Alone', 'MC', 150);

1 row created.

SQL > INSERT INTO Movie VALUES (5, 'A05', 'The Fugitive', 'MF', 200);

1 row created.

SQL > INSERT INTO Movie VALUES ('6', 'A03', 'Coma', 'MD', 100);

1 row created.

SQL > INSERT INTO Movie VALUES ('7', 'A02', 'Dracula', 'CD', 150);

1 row created.

SQL > INSERT INTO Movie VALUES (8, 'A06', 'Gone with the Wind', 'CB', 200);

1 row created.

SQL > INSERT INTO Movie VALUES (10, 'A05', 'Canyon Doctor', 'ZP', 100);

1 row created.

SQL > INSERT INTO Movies VALUES (4, 'A06', 'Home Alone', 'MC', 150);

1 row created.

SQL > INSERT INTO Movie VALUES (5, 'A05', 'The Fugitive', 'MF', 200);

1 row created.

SQL > INSERT INTO Movie VALUES (6, 'A03', 'Coma', 'MD', 100);

1 row created.

SQL > INSERT INTO Movie VALUES (7, 'A02', 'Dracula', 'Gd', 150);

1 row created.

SQL > INSERT INTO Movie VALUES (8, 'A06', 'Gone with the Wind', 'CB', 200);

1 row created.

SQL > INSERT INTO Movie VALUES (10, 'A05', 'Canyon Doctor', 'LP', 100);

1 row created.

Prove that entity integrity constraint is ensured by both the tables.

=> SQL> SELECT cust_id, COUNT(*)
FROM Customer
GROUP BY cust_id
HAVING COUNT(*) > 1;

no rows selected

SQL> SELECT *
FROM Customer
WHERE cust_id IS NULL;

no rows selected

SQL> SELECT MV_no, COUNT(*)
FROM Moviere
GROUP BY MV_no
HAVING COUNT(*) > 1;

no rows selected

SQL> SELECT *
FROM Movie
WHERE MV_no IS NULL;

no rows selected

2) Prove that entity integrity constraint is ensured by both the tables.

=> SQL> SELECT cust_id, COUNT(*)
FROM Customer
GROUP BY cust_id
HAVING COUNT(*) > 1;

no rows selected

SQL> SELECT *
FROM Customer
WHERE cust_id IS NULL;

no rows selected

SQL> SELECT MV_no, COUNT(*)
FROM Moviere
GROUP BY MV_no
HAVING COUNT(*) > 1;

no rows selected

SQL> SELECT *
FROM Movie
WHERE MV_no IS NULL;

no rows selected

3) SQL> SELECT cust_id
FROM Movie
WHERE cust_id NOT IN
(SELECT cust_id FROM Customer);

no rows Selected.

4) SQL> SELECT *
FROM Movie
WHERE Price < 100 OR Price >= 250;

no rows Selected

SQL> SELECT *
FROM Movie
WHERE Star IS NULL;

no rows Selected.

5) SQL> SELECT Title, Price
FROM Movie
WHERE Price > 100 AND Price < 200;

TITLE	PRICE
Bloody	181
Pretty Women	151
Home Alone	150
Dracula	150

3) SQL> SELECT cust_id
FROM Movie
WHERE cust_id NOT IN
(SELECT cust_id FROM customer);

no rows selected.

4) SQL> SELECT *
FROM Movie
WHERE Price < 100 OR Price >= 250;

no rows selected

SQL> SELECT *
FROM Movie
WHERE Star IS NULL;

no rows selected.

5) SQL> SELECT Title, Price
FROM Movie
WHERE Price > 100 AND Price < 200;

TITLE	PRICE
Bloody	181
Pretty Woman	151
Home Alone	150
Dracula	150

6)

```
SQL> SELECT DISTINCT Cust_id
  FROM Mark
  WHERE Stan IN ('JG', 'TC', 'MC');
```

CUS
A02
A04
A06

7)

```
SQL> SELECT *
  FROM Customer
  WHERE Area LIKE 'A%';
```

CUS	FNAME	LNAME	ARPHONE
A01	IVAN	Rose	SA6125467
A03	Pramada	Tangaste	DA 4360389
A04	Basu	Navindi	BA 6125401
A05	Ravi	Shridhar	NA

6) SQL> SELECT DISTINCT Cust-id
FROM Mark
WHERE Stan IN ('JG', 'TC', 'MC');

CUS
A02
A04
A06

7) SQL> SELECT *
FROM Customer
WHERE Area LIKE 'A%';

CUS	FNAME	LNAME	ARPHONE
A01	IVAN	Rose	SA6125467
A03	Pramada	Sangoshe	DA 4360389
A04	Basu	Navindi	BA 6125401
A05	Ravi	Shridhar	NA

8)

```
SQL> SELECT Title, Price
  FROM Movie
  WHERE Price <=180
  AND LENGTH(Title) =6;
no rows selected.
```

9)

```
SQL> SELECT Title,
       Price AS Original_Price,
       Price * 1.10 AS Incremented_Price
     FROM Movie;
```

TITLE	ORIGINAL_PRICE	INCREMENTED_PRICE
Blood	181	199.1
The Firm	200	220
Pretty Woman	151	166.1
Home Alone	150	165
The Fugitive	200	220
Coma	100	110
Dracula	150	165
Quick Change	100	110
Gone with the Wind	200	220
Carry on Doctor	100	110

10 rows selected.

8)

```
SQL> SELECT Title, Price
  FROM Movie
  WHERE Price <=180
  AND LENGTH(Title)=6;
no rows selected.
```

9)

```
SQL> SELECT Title,
       Price AS Original_Price,
       Price * 1.10 AS Incremented_Price
  FROM Movie;
```

TITLE	ORIGINAL_PRICE	INCREMENTED_PRICE
Blondy	181	199.1
The Firm	200	220
Pretty Woman	151	166.1
Home Alone	150	165
The Fugitive	200	220
Coma	100	110
Dracula	150	165
Quick Change	100	110
Gone with the Wind	200	220
Carry on Doctor	100	110

10 rows selected.

10) SQL> SELECT Fname || " " || Lname ||

' stays in ' || Area ||

' and his phone number is ' || Phone || '.' AS Customer_Deta

FROM Customer,

CUSTOMER_DETAILS

Ivan Ross stays in SA and his Phone number is 6125467.

Vandana Rao stays in MV and his Phone number is 556037.

Pramada Jagusta stays in DA and his Phone number is 6125401.

Basu Navidi stays in BA and his Phon number is 6125401.

Ravi Shridhar stays in NA and his Phone number is .

Rukmini A Iyer stays in 9H and his Phone number is 5125[illegible].

6 rows selected.

11) SQL> ALTER TABLE Customer

MODIFY Lname VARCHAR2(15) NOT NULL;

Table altered.

10)

```
SQL> SELECT Fname || " " || Lname ||
  'stays in' || Area ||
  'and his phone number is' || Phone || '.' AS Customer_Details
FROM Customer,
CUSTOMER_DETAILS
```

Ivan Ross stays in SA and his Phone number is 6125467.
Vandana Rao stays in MU and his Phone number is 5560379.
Pramada Jaguste stays in DA and his Phone number is 6125401
Basu Navidi stays in BA and his Phon number is 6125401.
Ravi Shridhar stays in NA and his Phone number is .
Rukmini Aiyer stays in RH and his Phone number is 5125279

6 rows selected.

11)

```
SQL> ALTER TABLE Customer
  MODIFY Lname VARCHAR2(15) NOT NULL;
```

Table altered.

12) SQL> SELECT Fname, Lname
FROM Customer
WHERE Phone IS NULL;

FName	LName
Ravi	Shridhar

13) SQL> UPDATE Customer
SET Phone = 9876543
WHERE Cust_id = 'A05';
1 row updated.

14) SQL> SELECT DISTINCT Cust_id
FROM Movie;

CUS
A02
A04
A01
A06
A05
A03

6 rows selected.

12) SQL> SELECT Fname, Lname
FROM Customer
WHERE Phone IS NULL;

FName	LName
Ravi	Shridhar

13) SQL> UPDATE Customer
SET Phone = 9876543
WHERE Cust_id = 'A05';
1 row updated.

14) SQL> SELECT DISTINCT Cust_id
FROM Movie;

CUS
A02
A04
A01
A06
A05
A03

6 rows selected.

15) SQL> ALTER TABLE Movie
MODIFY Star VARCHAR2(2) NULL;
Table altered.

16) SQL> DELETE FROM Customer
WHERE Cust_Id = A061;
DELETE FROM Customer
*
ERROR at line 1:
ORA-02292: integrity constraint (SYSTEM.SYS_C008329)
violated - child record found.

17) SQL> DELETE FROM Movie
WHERE MV_NO = 10;
3 row deleted.

18) SQL> DROP TABLE Customer;
DROP TABLE Customer
*
ERROR at line 1:
ORA-02449: Unique/Primary keys in table referenced by
foreign keys.

15) SQL> ALTER TABLE Movie
MODIFY Star VARCHAR2(2) NULL;
Table altered.

16) SQL> DELETE FROM Customer
WHERE Cust_id = A061;
DELETE FROM Customer
*
ERROR at line 1:
ORA-02292: integrity constraint (SYSTEM.SYS_C008324
violated - child record found.

17) SQL> DELETE FROM Movie
WHERE MV_NO = 10;
3 row deleted.

18) SQL> DROP TABLE Customer;
DROP TABLE Customer
*
ERROR at line 1:
ORA-02449: Unique/Primary keys in table referenced by
foreign keys.

1) SQL> DROP TABLE Movie;

Table dropped.

2) SQL> SELECT Constraint_Name
FROM User_Constraints
WHERE Table_Name = 'Movie'
AND Constraint_Type = 'R';

no rows selected.